Friedel oscillations (Polarization operator carefully done)

$$P^{(1)}(\bar{q},\omega) = \chi_0(\bar{q},\omega) = \underset{\text{spin}}{(2)}\int \frac{d\vec{k}}{(2\pi)^3} \frac{f(\xi_k) - f(\xi_{k+q})}{\omega - \xi_{k+q} + \xi_k + i\delta} =$$

$$\xi_{k+q} - \xi_k = \frac{(\bar{k}+\bar{q})^2 - k^2}{2m} = \frac{\vec{k}\cdot\vec{q}}{m} + \frac{q^2}{2m}$$

$$\left(\begin{array}{c}\bar{k}\to\bar{k}-\bar{q}\\ \text{in } 2^{nd}\\ \text{term}\end{array}\right) = 2\int \frac{d^3\vec{k}}{(2\pi)^2}\left\{\frac{f(\xi_k)}{\omega - \frac{q^2}{2m} - \frac{\bar{k}\cdot\bar{q}}{m} + i\delta} - \frac{f(\xi_k)}{\omega + \frac{q^2}{2m} - \frac{\bar{k}\cdot\bar{q}}{m} + i\delta}\right\}$$

$$\xi_{k+q} - \xi_k \to \xi_k - \xi_{k-q} = \frac{k^2 - (\bar{k}-\bar{q})^2}{2m} = \frac{\bar{k}\cdot\bar{q}}{m} - \frac{q^2}{2m}$$

Observe that if we choose to parameterize

$\vec{k} = k_{\parallel}\hat{q} + k_1\hat{e}_1 + k_2\hat{e}_2$ where $\hat{q}$ = unit vector along $\vec{q}$

$\hat{e}_{1,2}$ = two orthogonal to $\hat{q}$ unit vectors

i.e. can set

$\vec{k} = k_x\hat{e}_x + k_y\hat{e}_y + k_z\hat{e}_z$ and $\vec{q} = q\,\hat{e}_x$.

$\Rightarrow \vec{k}\cdot\vec{q} = q\,k_x$

See that $k_{y,z}$ enter only via $f(\xi_k)$.

At $T=0$ $\quad f(\xi_k) = \theta(-\xi_k) = \theta(\mu - \varepsilon_k) = \theta\left(\frac{k_F^2 - k^2}{2m}\right)$

$$\vec{k}^2 = k_x^2 + k_y^2 + k_z^2 = k_x^2 + \vec{k}_\perp^2, \quad \vec{k}_\perp = (0, k_y, k_z)$$

$$\chi_0(\vec{q},\omega) = 2\int\frac{dk_x}{(2\pi)}\int\frac{dk_y\,dk_z}{(2\pi)^2}\,\theta(k_F^2 - k_x^2 - \vec{k}_\perp^2)\cdot$$

$$\cdot\left\{\frac{1}{\omega - \frac{q^2}{2m} - \frac{qk_x}{m} + i\delta} - \frac{1}{\omega + \frac{q^2}{2m} - \frac{qk_x}{m} + i\delta}\right\}$$

$$\int dk_y\, dk_z = \int_0^\infty dk_\perp\, k_\perp \int_0^{2\pi} d\varphi = \pi \int_0^\infty d(k_\perp^2)$$

$$\int_0^\infty d(k_\perp^2)\quad \theta\left(k_F^2 - k_x^2 - k_\perp^2\right) = \int_0^{k_F^2-k_x^2} dk_\perp^2 \cdot 1 = k_F^2 - k_x^2$$

$$0 \le k_\perp^2 \le k_F^2 - k_x^2 \quad \text{and} \quad k_x^2 \le k_F^2$$

$$\Rightarrow \chi_0(\vec{q},\omega) = \frac{2\pi}{(2\pi)^3} \int_{-k_F}^{k_F} dk_x \left(k_F^2 - k_x^2\right) \left\{ \frac{1}{\omega - \frac{q^2}{2m} - \frac{q k_x}{m} + i\delta} - \frac{1}{\omega + \frac{q^2}{2m} - \frac{q k_x}{m} + i\delta} \right\}$$

For simplicity, consider $\chi_0(\vec{q},\omega=0) = \chi_0'(\vec{q},0)$ only (more general calculation with $\omega \neq 0$ is easily possible).

$$\chi_0(\vec{q},0) = \frac{-1}{(2\pi)^2} \int_{-k_F}^{k_F} dk_x \left(k_F^2 - k_x^2\right) \left\{ \frac{1}{\frac{q^2}{2m} + \frac{q k_x}{m} + i\delta} + \frac{1}{\frac{q^2}{2m} - \frac{q k_x}{m} + i\delta} \right\}$$

The presence of $i\delta$ means that integral should be understood as Principal Value one.

$$= \frac{-1}{(2\pi)^2} \frac{q^2}{m} \int_{-k_F}^{k_F} dk_x \frac{k_F^2 - k_x^2}{\left(\frac{q^2}{2m}\right)^2 - \left(\frac{q k_x}{m}\right)^2} =$$

$$k_x = k_F \cdot x$$

$$= \frac{-1}{(2\pi)^2} \frac{q^2 k_F}{m} \int_{-1}^{1} dx \frac{(1-x^2) k_F^2}{\frac{q^4}{4m^2} - \frac{q^2 k_F^2}{m^2} x^2} =$$

$$= \frac{-1}{(2\pi)^2} \frac{q^2 k_F^3}{m} \frac{m^2}{q^2 k_F^2} \int_{-1}^{1} dx \frac{1-x^2}{\left(\frac{q}{2k_F}\right)^2 - x^2}$$

even function of x

$$\text{let } a = \left|\frac{q}{2k_F}\right| > 0$$

Need $\mathcal{J} = \text{P.V.}\, 2\int_0^1 dx \, \frac{1-x^2}{a^2-x^2} = 2\int_0^1 dx \, \frac{a^2-x^2+1-a^2}{a^2-x^2} =$

$$= 2 + 2\cdot\frac{1-a^2}{2a}\,\text{P.V.}\int_0^1 dx \left(\frac{1}{a-x} + \frac{1}{a+x}\right)$$

Assume $0 < a < 1$: $(1-a>0,\ a-1<0)$

$$\mathcal{J}_1 = \text{P.V.}\int_0^1 dx \left(\frac{1}{a-x} + \frac{1}{a+x}\right) =$$

$$= \int_0^a dx \left(\frac{1}{a-x} + \frac{1}{a+x}\right) + \int_a^1 dx \left(\frac{1}{a-x} + \frac{1}{a+x}\right)$$

$$= \Big(\ln(a+x) - \ln(a-x)\Big)\Big|_0^a + \Big(-\ln(x-a) + \ln(a+x)\Big)\Big|_a^1$$

$$= \ln\frac{a+x}{a-x}\Big|_0^a + \ln\frac{a+x}{x-a}\Big|_a^1 =$$

$$\cancel{= \ln\frac{2a}{0} - \ln\frac{a-1}{a+1} + \ln\frac{a+1}{1-a} - \ln\frac{2a}{0}}$$

$$= \ln\frac{2a}{0} - \ln\frac{a}{a} + \ln\frac{a+1}{1-a} - \ln\frac{2a}{0} = \ln\frac{1+a}{1-a}$$

for $0<a<1$.

Assume $a>1$:

$$\mathcal{J}_1 = \text{P.V.}\int_0^1 dx \left(\frac{1}{a+x} + \frac{1}{a-x}\right) =$$

$$= \Big(\ln(a+x) - \ln(a-x)\Big)\Big|_0^1 = \ln\frac{a+1}{a-1} \quad \text{for } a>1.$$

$\Rightarrow \mathcal{J}_1 = \ln\left|\frac{a+1}{a-1}\right|$ for general a.

$$\Rightarrow \mathcal{J} = 2 + \frac{1-a^2}{a}\ln\left|\frac{a+1}{a-1}\right|.$$

$$\chi_0(\vec{q},0) = -\frac{m k_F}{4\pi^2}\left\{2 + \left(\frac{2k_F}{q} - \frac{q}{2k_F}\right)\ln\left|\frac{2k_F+q}{2k_F-q}\right|\right\}$$

let $q = 2k_F(1-x)$, $x = 1 - \frac{q}{2k_F}$

$$\Rightarrow \{\ \} = 2 + \left(\frac{1}{1-x} - (1-x)\right)\ln\left|\frac{2-x}{x}\right| = (x \ll 1) =$$

$$\approx 2 + 2x\ln\frac{2}{|x|} \quad \text{for } x \ll 1.$$

Thus $$\chi_0(\vec{q},0) \cong -\frac{m k_F}{2\pi^2}\left(1 + \left(1 - \frac{q}{2k_F}\right)\ln\left|\frac{4k_F}{2k_F-q}\right|\right)$$

for $\frac{q}{2k_F} = 1 - x \approx 1$.

* Derivative of this function $\sim \ln\frac{1}{|x|}$ for $x \ll 1$!

diverges as

Responsible for power-law decay.

For $d=3$ Fermi gas density of states on the fermi level is $N_0 = \nu = \frac{m k_F}{2\pi^2} \Rightarrow \frac{m k_F}{2\pi^2} = \nu$.

$$\chi_0(\vec{q},0) = P^{(1)}(q,0) = -\frac{1}{2}\nu\left\{2 + \left(\frac{2k_F}{q} - \frac{q}{2k_F}\right)\ln\left|\frac{2k_F+q}{2k_F-q}\right|\right\}.$$

Let us look on induced (screening) charge density again.

$$\frac{\rho_s(\vec{q},\omega)}{\rho_{ext}(\vec{q},\omega)} = \frac{1}{\varepsilon(\vec{q},\omega)} - 1 = \frac{1}{1 - V(\vec{q})\, P_R^{(1)}(\vec{q},\omega)} - 1$$

Let ρ_{ext} to describe static point charge Q_{ext}.

$\Rightarrow \quad \rho_{ext}(\vec{r},t) = Q_{ext}\, \delta(\vec{r})$

$\Rightarrow \rho_{ext}(\vec{q},\omega) = Q_{ext}\, \delta(\omega)$ (~~[illegible]~~ static impurity).

$$\Rightarrow \underbrace{\rho_s(\vec{q},\omega=0)}_{\text{write it as } \rho_s(\vec{q})} = \left(\frac{1}{\varepsilon(\vec{q},0)} - 1\right) Q_{ext}.$$

Then induced charge density in <u>real space</u> is

$$\rho_s(\vec{r}) = \int \frac{d^3\vec{q}}{(2\pi)^3}\, e^{i\vec{q}\cdot\vec{r}}\, \rho_s(\vec{q}) =$$

$$= Q_{ext} \int \frac{d\vec{q}}{(2\pi)^3} \left(\frac{1}{\varepsilon(\vec{q},0)} - 1\right) e^{i\vec{q}\cdot\vec{r}}$$

The (-1) term gives $(-)\int \frac{d\vec{q}}{(2\pi)^3}\, e^{i\vec{q}\cdot\vec{r}} = (-)\,\delta(\vec{r})$

and will be omitted in the following as we are interested in long-distance behavior of $\rho_s(\vec{r})$.

$$\frac{\rho_s(\vec{r})}{Q_{ext}} = \int \frac{d^3\vec{q}}{(2\pi)^3} e^{i\vec{q}\cdot\vec{r}} \left(\frac{1}{\varepsilon(q,0)} - 1\right)$$

Since $\varepsilon(q,0)$ depends only on magnitude of $\vec{q}$, integrate over angle: $\int \frac{d\vec{q}}{(2\pi)^3} e^{i\vec{q}\cdot\vec{r}} = \frac{2\pi}{(2\pi)^3} \int_0^\infty dq\, q^2 \underbrace{\int_0^\pi d\theta \sin\theta\, e^{iqr\cos\theta}}_{\int_{-1}^{1} d(\cos\theta)}$

$$= \frac{1}{(2\pi)^2} \int_0^\infty \frac{dq\, q^2}{iqr} \left(e^{iqr} - e^{-iqr}\right) =$$

$$= \frac{1}{2\pi^2 r} \int_0^\infty dq\, q \sin(qr)$$

$$\Rightarrow \frac{\rho_s(\vec{r})}{Q_{ext}} = \frac{1}{2\pi^2 r} \int_0^\infty dq\, q \frac{\sin qr}{\varepsilon(q,0)}$$

$\boxed{\text{Need to extract the leading term for } r \to \infty}$

$$\varepsilon(q,0) = 1 - \underset{\frac{4\pi e^2}{q^2}}{\underset{\parallel}{V(\vec{q})}} P^{(1)}(q,0) = 1 + \frac{2\pi e^2 \nu}{q^2} \left\{ 2 + \left(\frac{2k_F}{q} - \frac{q}{2k_F}\right) \ln\left|\frac{2k_F+q}{2k_F-q}\right| \right\}$$

$P^{(1)}(q,0)$ has logarithmically divergent derivative for $q \approx 2k_F \Rightarrow \varepsilon(q,0)$ has the same property.

Need to understand how singular derivative affects the result $\Rightarrow$ to expose it, integrate by parts!

let $J = \int_0^\infty dq\, f(q) \sin qr = \int_0^\infty d\left(-\frac{\cos qr}{r}\right) f(q) =$

$$= -\frac{1}{r} \cos qr\, \cancel{f(q)}\Big|_0^\infty + \frac{1}{r} \int_0^\infty dq \cos qr\, f'(q)$$

$$= \frac{1}{r} \int_0^\infty d\left(\frac{\sin qr}{r}\right) f'(q) = \frac{1}{r^2} \sin qr\, f'(q)\Big|_0^\infty - \frac{1}{r^2} \int_0^\infty dq \sin(qr) f''(q)$$

$$\Rightarrow \rho_s(r) = -\frac{Q_{ext}}{2\pi^2 r^3}\int_0^\infty dq\ \sin(qr)\cdot\frac{d^2}{dq^2}\left(\frac{q}{\varepsilon(q,0)}\right)$$

We found for $q = 2k_F(1-x)$, $x = 1-\frac{q}{2k_F}$ and <u>x small</u>

$$P^{(1)}(q,0) \approx -\nu\left(1 + x\ln\left|\frac{2}{x}\right|\right)$$

$$\Rightarrow \varepsilon(q\approx 2k_F,0) \cong 1 + \frac{4\pi e^2\nu}{q^2}\left(1 + x\ln\frac{2}{|x|}\right) =$$ replace $q=2k_F$ in all non-singular parts

$$= 1 + \frac{4\pi e^2\nu}{(2k_F)^2}\left(1 + x\ln\frac{2}{|x|}\right)$$

$$\frac{d}{dq} = \frac{d}{dx}\frac{dx}{dq} = \frac{-1}{2k_F}\frac{d}{dx}$$

$$\rho_s(r) = \left(-\frac{Q_{ext}}{2\pi^2 r^3}\right)\int_1^{-\infty}(-2k_F)\,dx\ \sin\left(2k_F r(1-x)\right)\frac{1}{(2k_F)^2}\cdot$$

$$\cdot\frac{d^2}{dx^2}\frac{2k_F}{1+\frac{\pi e^2\nu}{k_F^2}\left(1+x\ln\frac{2}{|x|}\right)}$$

$$= \frac{Q_{ext}}{2\pi^2 r^3}\int_1^{-\infty}dx\ \left\{\cos(2k_F r x)\sin(2k_F r) - \cos(2k_F r)\sin(2k_F r x)\right\}$$

$$\cdot\frac{d^2}{dx^2}\left(\frac{1}{1+A+Ax\ln\frac{2}{|x|}}\right)$$

where $A = \frac{\pi e^2\nu}{k_F^2} = \frac{\pi e^2}{k_F^2}\frac{mk_F}{2\pi^2} = \frac{e^2 m}{2\pi k_F} = \frac{1}{2\pi}\frac{e^2}{v_F} \ll 1$ for usual conductors

Taking the derivative and extracting the leading behavior (in $x\to 0$) using Mathematica, we find

$$\frac{d^2}{dx^2}\left(\quad\right) \to \frac{A}{(1+A)^2 x}$$

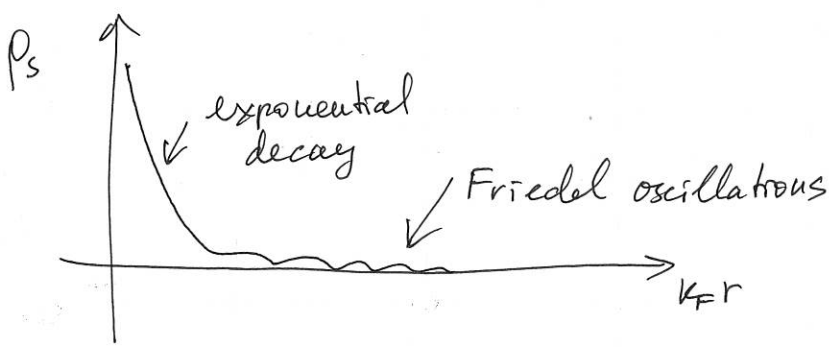
Due to convergence (see below), the integration can be extended to $\to -\int_{-\infty}^{\infty}dx\ldots$

Then we only need to calculate ($a = 2k_F r > 0$ here)

$$\int_{-\infty}^{\infty} dx \, \frac{\cos ax}{x} = 0 \quad \text{but} \quad \int_{-\infty}^{\infty} dx \, \frac{\sin ax}{x} = \pi \, \mathrm{Sign}(a)$$

$$\Rightarrow \rho_s(r) = -\frac{Q_{ext}}{2\pi^2 r^3} \frac{A}{(1+A)^2} \left\{ \int_{-\infty}^{\infty} dx \, \frac{\sin(2k_F r)\cos(2k_F r x)}{x} - \frac{\cos(2k_F r)\sin(2k_F r x)}{x} \right\} = \frac{Q_{ext}}{2\pi r^3} \frac{A\cos(2k_F r)}{(1+A)^2}$$

$\Rightarrow$ density of screening charge decays as $\frac{\cos(2k_F r)}{r^3}$ far away from the impurity. } Friedel oscillations!!!



Plot and compare $P^{(1)}(q,0)$ in different dimensions.